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Education to Eternity

JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of CSE/AI
"A place to learn; A chance to grow"
2025-2026 (Odd Semester)

MSE-1 EXAMINATION

Course: BTech in CSE/AI **Year/Semester:** 3rd Semester (2nd Year)
Subject: Numerical Technique and Discrete Mathematics
Subject Code: CS3M001
Max Marks: 20 **Date:** 11/08/2025 **Duration: - 1 Hr.**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO 1: Define fundamental concepts of linear algebra, numerical methods, probability, and Group Theory.

CO 2 Illustrate the concepts of matrices, numerical techniques, Group Theory methods and probability distributions with relevant examples.

CO 3: Apply linear algebra techniques, numerical methods, and Probability and Group Theory to solve engineering mathematical problems.

CO 4: Analyze the concept of probability distributions, numerical technique, Group Theory, and linear algebra technique.

CO 5: Evaluate the mathematical problem by using, Eigen values and Eigen vectors, Sylvester's theorem, Bisection method, Newton-Raphson method and their convergences. Solution of system of linear equations by Gauss Seidel method, Gauss elimination method, Crout's method. Mathematical Expectation, probability and Group Theory.

CO 6: Develop Mathematical Models based on linear algebra and probability distributions & Group Theory to solve real-life and engineering-related Problems.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Find the Eigen Value of Matrix $A=\begin{bmatrix}1 & 0\\0 & 3\end{bmatrix}$ (a) 2,3 (b) 1,3 (c) 0,1 (d) 1,1	L1/CO1	1
Q.2	What is the Rank of Matrix $A=\begin{bmatrix}1 & 2 & 3\\0 & 1 & 1\\0 & 2 & 2\end{bmatrix}$ (a) 0 (b) 1 (c) 3 (d) 2	L1/CO1	1
Q.3	Find the Adjoint of given Matrix $A=\begin{bmatrix}5 & 4\\-2 & -3\end{bmatrix}$ (a) $\begin{bmatrix}5 & 4\\-2 & -3\end{bmatrix}$ (b) $\begin{bmatrix}-3 & -4\\2 & 5\end{bmatrix}$ (c) $\begin{bmatrix}5 & -4\\2 & -3\end{bmatrix}$ (d) $\begin{bmatrix}-3 & -4\\2 & -3\end{bmatrix}$	L1/CO1	1

Q.4 Tell the Remaining Entries of the Symmetric Matrix, Where $a_{11} = 1$ $a_{22} = 2$, $a_{33} = 3$, $a_{12} = 2$, $a_{13} = 3$, $a_{23} = 4$ Find a_{21}, a_{31}, a_{32} L1/CO1 1

(a) 2,3,-4 (b) -2,3,4 (c) 2,3,4 (d) -2,-3,-4

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief.**

Q.5 Illustrate the root of equation $xe^x = 1$ in the interval [0,1] by using Bisection Method upto two Iteration. L2/CO2 2

Q.6 Interpret the root of the equation $3x - cosx - 1 = 0$ in the Interval [0.6, 0.7] by using Newton-Raphson Method. L2/CO2 2

Q.7 Interpret the quadratic form for the corresponding Matrix.

$$A=\begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{bmatrix}$$ L2/CO2 2

Q.8 Solve the Eigen value for the given Matrix.

$$A=\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$ L3/CO3 2

SECTION C

1. Solve any two questions.

Q.9 Test the Consistency and solve; L4/CO4 4

$5x + 3y + 7z = 4;\ \ 3x + 26y + 2z = 9;\ \ 7x + 2y + 10z = 5$

Q.10 Apply Gauss Seidal Method to solve the equation; L3/CO3 4

$2x - 3y + 20z = 25;\ \ 20x + y - 2z = 17;\ \ 3x + 20y - z = -18$

Q.11 Analyse by Gauss Elimination Method L4/CO4 4

$2x + 4y - 6z = -4;\ x + 5y + 3z = 10;\ x + 3y + 2z = 5.$